

2015-2016 – Econ590
Topics in Macroeconomics

Lecture 0 : Business Cycle Facts and Detrending
Supplement : Main elements to retain about
spectral analysis and filtering

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Main elements

- ▶ Given an autocovariance function $\gamma(\tau)$, the (power) spectrum is defined as (the Fourier transform of the auto-covariance function)

$$s(\omega) = \frac{f(\omega)}{2\pi} = \frac{1}{2\pi} \sum_{\tau=-\infty}^{\infty} \gamma(\tau) \exp^{i\omega\tau}$$

or can be rewritten as (since $\exp^{iz} = \cos(z) + i \sin(z)$, where $i = \sqrt{-1}$)

$$s(\omega) = \frac{1}{2\pi} [\gamma(0) + 2 \sum_{\tau=1}^{\infty} \gamma(\tau) \cos(\omega\tau)]$$

- ▶ given $s(\omega)$ one can recover the autocovariance function using

$$\gamma(\tau) = \int_{-\pi}^{\pi} s(\omega) \exp^{i\omega\tau} d\omega$$

- ▶ this implies that $s(\omega)$ contains the same information as $\gamma(\tau)$

- ▶ What is interesting about $s(\omega)$?
- ▶ $s(\omega)$ give the the importance of fluctuations of frequency ω of the time series ($\int_0^z s(\omega)d\omega$ gives the amount of variance of a time series attributable to cycles with frequencies between 0 and z), where ω is measured as the number of radians of the wave in a unit of time.
- ▶ The interpretation of spectrum can seen by considering the spectral representation of the time series

$$x_t = \int_0^\pi a(\omega)\cos(\omega t)d\omega + \int_0^\pi b(\omega)\sin(\omega t)d\omega$$
- ▶ where a and b are seen as mean zero random variables, with variance $2\pi s(\omega)$.
- ▶ The period associated with ω , in units of data intervals, $\frac{2\pi}{\omega}$. If data is in quarters, and $\omega = .5\pi$, this corresponds to a year.

- ▶ The spectrum is sometimes plotted directly as a function of ω , over the range $[0, \pi]$, or in units of 2π over the range $[0, .5]$.
- ▶ The later allows for an easy interpretation in term of of periods as $\frac{1}{N}$, where N is the number of 2π . (N would be the number on the x-axis in this case). For example, if data is in quarters, .25 is 4 quarter and hence a year.

- The spectral representation is particularly helpful when considering the effects of a moving average filter. For example, suppose,

$$x_t^f = C(L)x_t$$

- where L is the lag operator and $C(L) = \sum_{j=0}^{\infty} c_j L^j$, and x_t is the unfiltered times series under consideration. If we denote by $G(\omega)$ the gain of this filter, then we have

$$s^f(\omega) = G(\omega)^2 s(\omega)$$

- where the gain is calculated as

$$G(\omega) = \sqrt{C(\exp^{-i\omega})C(\exp^{i\omega})}$$

- Hence, we can see a moving average filter as changing the emphasis of different frequencies. (mention the Slutsky (?) observation as a caution about the spurious creation of cycles). Note : the inverse can also be done. Given a gain, one can find $C(L)$.